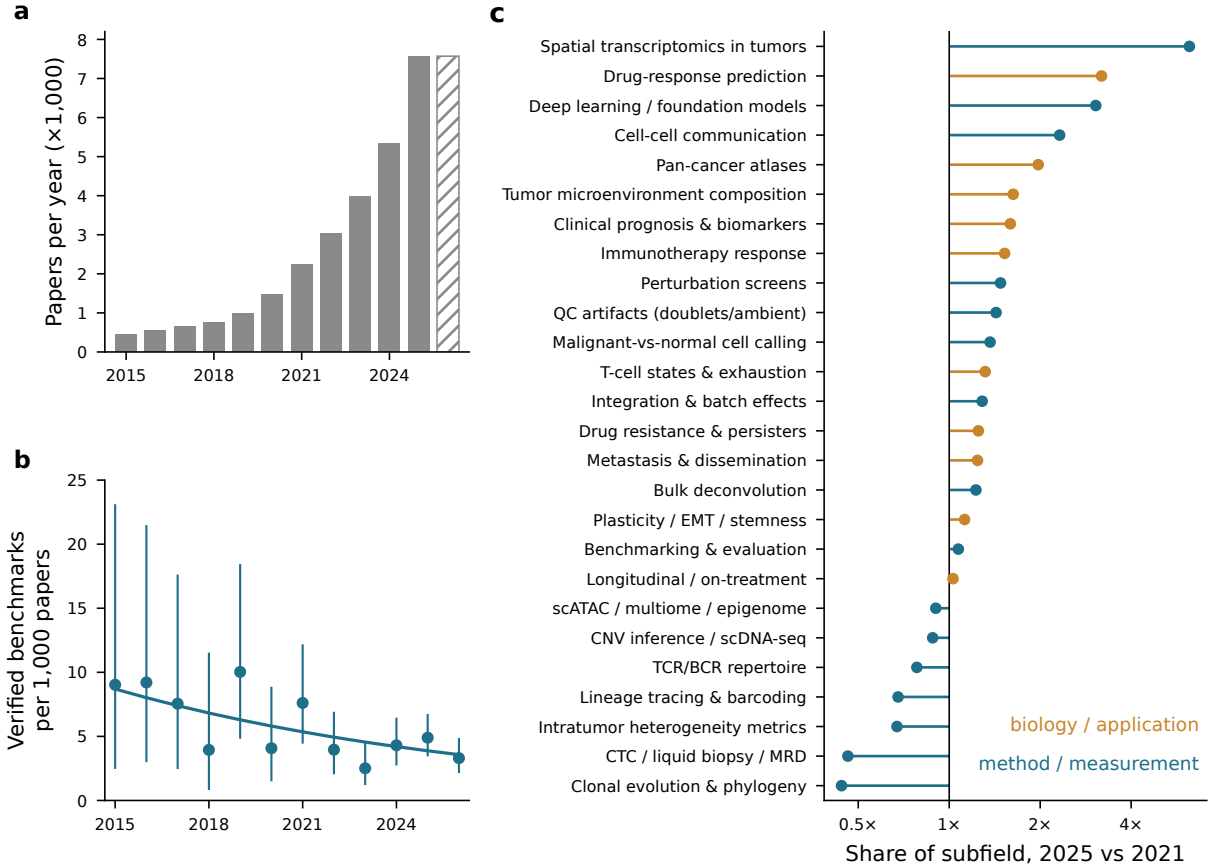
a
Papers per year (×1,000)
0
1
2
3
4
5
6
7
8
2015
2018
2021
2024
b
Verified benchmarks per 1,000 papers
0
5
10
15
20
25
2015
2018
2021
2024
c
Spatial transcriptomics in tumors
Drug-response prediction
Deep learning / foundation models
Cell-cell communication
Pan-cancer atlases
Tumor microenvironment composition
Clinical prognosis & biomarkers
Immunotherapy response
Perturbation screens
QC artifacts (doublets/ambient)
Malignant-vs-normal cell calling
T-cell states & exhaustion
Integration & batch effects
Drug resistance & persisters
Metastasis & dissemination
Bulk deconvolution
Plasticity / EMT / stemness
Benchmarking & evaluation
Longitudinal / on-treatment
scATAC / multiome / epigenome
CNV inference / scDNA-seq
TCR/BCR repertoire
Lineage tracing & barcoding
Intratumor heterogeneity metrics
CTC / liquid biopsy / MRD
Clonal evolution & phylogeny
biology / application
method / measurement
0.5×
1×
2×
4×
Share of subfield, 2025 vs 2021